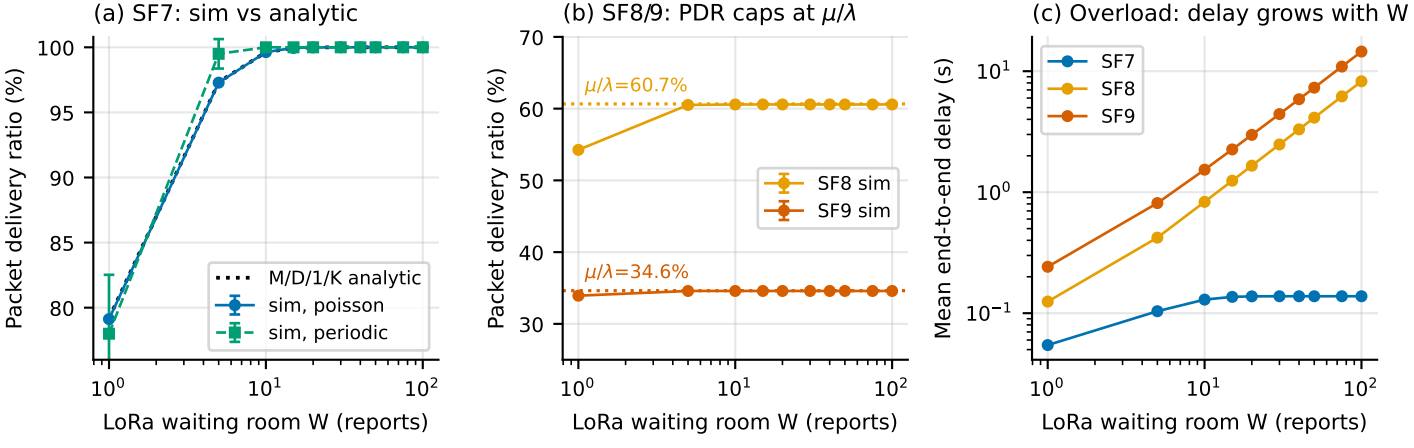
(a) SF7: sim vs analytic
Packet delivery ratio (%)
100
95
90
85
80
10^0
10^1
10^2
LoRa waiting room W (reports)
M/D/1/K analytic
sim, poisson
sim, periodic
(b) SF8/9: PDR caps at μ/λ
Packet delivery ratio (%)
70
60
50
40
30
μ/λ=60.7%
μ/λ=34.6%
SF8 sim
SF9 sim
10^0
10^1
10^2
LoRa waiting room W (reports)
(c) Overload: delay grows with W
Mean end-to-end delay (s)
10^1
10^0
10^{-1}
SF7
SF8
SF9
10^0
10^1
10^2
LoRa waiting room W (reports)